SUPPORTING ADULT GUIDANCE

Reading real parabolas

Quadratics. For any supporting adult — teacher, practitioner, parent or carer. **You do not need to be a mathematics specialist to support this lesson.**

What this lesson does

The unit's **applied finale**. Its single job is **interpretation**: reading a real quadratic graph and saying what its **turning point** means in context. Two ideas do all the work. First, **maximum vs minimum**: a curve that opens **downward** (∩, a negative leading coefficient) has a **maximum** at its turning point; one that opens **upward** (U) has a **minimum**. Second, **the turning point is an answer**: its coordinates tell you the most or least of something and exactly when or where it happens — a ball's greatest height and the time it occurs, the price that maximises revenue, the output that minimises cost. The lesson also includes an honest optimisation: for a **fixed perimeter**, the enclosed area is a downward quadratic, maximised by a **square**. No new algebra is introduced; the aim is to connect everything the unit has built to situations that matter.

Driving Question: *Where do U-shaped curves show up, and what do their turning points mean?*

The mathematics, in plain terms

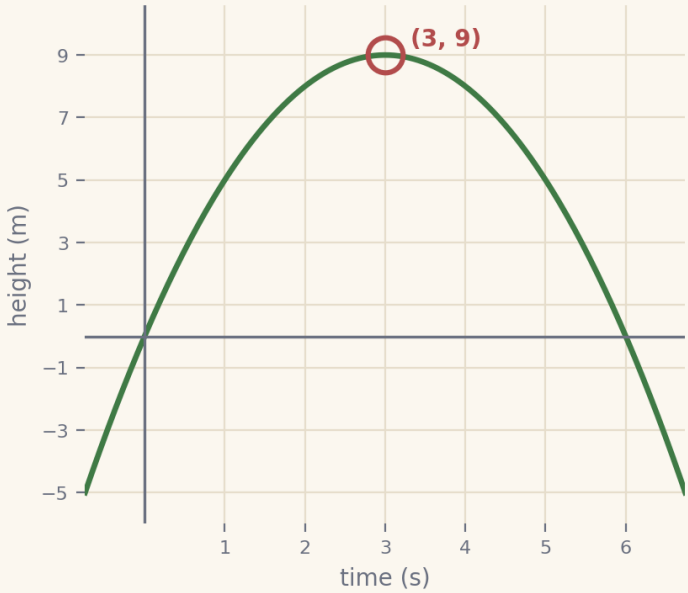
A **parabola** is not just a shape in a maths book — it is the path of a thrown ball, the arc of a fountain, and the graph of many real relationships (revenue against price, cost against quantity, area against a side length). The most useful part of such a graph is its **turning point**. If the curve opens **downward** (∩) the turning point is the **highest** point — a **maximum**; if it opens **upward** (U) it is the **lowest** — a **minimum**. Reading it is just reading coordinates from **labelled axes**: go to the very top (or bottom) of the curve, read straight down for the first number and across for the second. Then say what those numbers *mean*: on a height–time graph, a turning point at (3, 9) means "the greatest height is 9 metres, reached at 3 seconds". The most valuable support is to keep asking "what does this point mean in the story?" — turning a pair of numbers back into a real answer. The fencing task makes the same idea concrete: with a set length of fence, a **square** encloses the most growing area.

The worked method the learner sees

These are the two worked examples the learner meets on screen, with the lesson's exact method and notation. They are here for consistency, not because the mathematics is demanding — whether you teach mathematics or are meeting it afresh alongside the learner, working from the same steps and the same language keeps your support aligned with what is on the screen, and lets you point back to a line the learner has already seen. Skip them freely if the method is already familiar.

Example 1 — the greatest height

A ball's height is $h = -(t - 3)^2 + 9$ (metres, against time in seconds). Find its greatest height and when it happens.

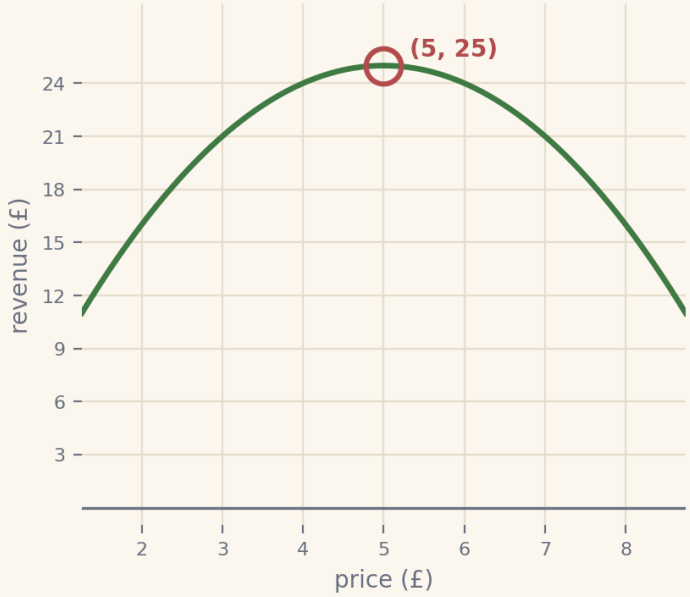


1. the curve opens	downward (∩), so the turning point is a maximum
2. the top is where the bracket is 0	$t - 3 = 0$, so $t = 3$
3. the height there	$h = 0 + 9 = 9$
4. so, in words	greatest height 9 m, reached at 3 s

the ball reaches a maximum height of 9 m at 3 seconds

Example 2 — the best price for the most revenue

A shop's revenue is R = –(price – 5)^2 + 25 (£, against price in £). Find the price that earns the most, and how much.



1. the curve opens	downward (∩), so the turning point is a maximum
2. the top is where the bracket is 0	price – 5 = 0, so price = 5
3. the revenue there	R = 0 + 25 = 25
4. so, in words	most revenue £25, at a price of £5

a price of £5 earns the most revenue, £25

Read the turning point, then say what it means

Two moves carry this lesson. First, **classify**: does the curve open up or down? Down (∩) means the turning point is a **maximum** (a greatest value); up (U) means a **minimum** (a least value). Tie this to the context — greatest height and most revenue are maxima; least cost and smallest depth are minima. Second, **read and interpret**: read the turning point's coordinates from the labelled axes, then translate them back into the situation — the most or least of something, and when or where it happens. Insist on the units and the meaning, not just the numbers: "9 metres, at 3 seconds", not "(3, 9)". The fencing task is a gentle optimisation — moving the side length and watching the area rise to a maximum at the square — and a genuine food-growing context. No new technique is needed here; the work is entirely in *connecting* the unit's mathematics to the world.

The shape of the lesson

Timings are invitations, not deadlines. A learner may need much longer on one phase and skim another — that is expected. Each Cornerstone is given an honest mathematical role here:

Cornerstone	Its role here	Invitational timing
Connection	The shape is everywhere: read the top of a thrown ball's height–time arc.	~8 min
Movement	Maximum or minimum?: decide which way each real curve opens and what that means.	~9 min
Reflection	The parabola in the world: read a turning point from labelled axes and interpret it.	~12 min
Creativity	The biggest growing space: find the side length that maximises the area for a fixed fence.	~12 min
Rest	A pause after seeing the quadratic's four faces become one.	~2 min
Nutrition	Mode A (contextual): a genuine food-growing optimisation — a square encloses the most area for a fixed fence.	~2 min

Common misconceptions, and gentle repairs

These are the sticking points to expect. In every case, the aim is a question that returns the thinking to the learner — never to supply the answer.

Thinking a downward curve has a minimum

Repair: “A ∩ opens downward, so its turning point is the highest point — a maximum. A minimum is the bottom of a U.”

Reading only one coordinate of the turning point

Repair: “A turning point has two parts: the value (how much) and where it happens. (3, 9) means 9 m AND at 3 s.”

Giving coordinates without meaning or units

Repair: “Translate the point back into the story: not '(5, 25)' but 'the most revenue, £25, at a price of £5'.”

Assuming a longer, thinner bed gives more area

Repair: “Try it: with 20 m of fence, 2 × 8 gives 16 m^2 but 5 × 5 gives 25 m^2. The square encloses the most.”

Confusing where a curve crosses zero with its turning point

Repair: “The zeros are where the ball leaves and lands; the turning point is the top of the arc — the greatest height.”

Protecting the support pathway

The lesson offers support in a deliberate order: **Socratic prompt** → **first try** → **hint** → **second try** → **worked solution**. The worked solution for each question stays locked until the learner has used the prompt, used the hint, and checked two answers. This is by design: the steps *are* the learning.

Please hold this line

Do not read the worked solution aloud, or talk a learner straight to the answer, while it is still locked. If they are stuck, use the Reconnection Routes, the prompt and the hint, or ask one of the repair questions above. Arriving at a correct line of reasoning — even slowly — is worth far more than being handed the result.

Reconnection Routes

If an earlier idea is blocking progress, the lesson’s Reconnection Routes offer short recaps of exactly the prerequisites this lesson needs: the parabola and its turning point (Lessons 5–7), reading coordinates from labelled axes, and substituting into a simple expression. A learner can choose one independently, or you can invite one without requiring it. Using a route is part of learning, not evidence of falling behind, and it always returns the learner to their place.

Supporting through questions: an example

Keep the learner reading turning points and saying what they mean:

- “Does this curve open up or down? So is the turning point a maximum or a minimum?”
- “Read the turning point. What is the value, and when or where does it happen?”
- “Say it as a sentence with units — the most/least of what, and when?”
- “With 20 m of fence, what area does w = 3 give? And w = 5? Which is bigger?”

If they stall, that is the moment for a Reconnection Route or the lesson’s prompt and hint — not for the answer.

Questions you might have

“How do I tell a maximum from a minimum?”

Look at which way the curve opens. Downward (∩) means the turning point is the highest point — a maximum. Upward (U) means it is the lowest — a minimum.

“What does the turning point actually tell you?”

Both of its coordinates matter: one is the most or least of something (a value), the other is when or where it happens. Reading a graph well means saying both, with units.

“Why does a square give the most area for a fixed fence?”

Because the area, written against one side length, is a downward parabola, and its maximum falls exactly where the two sides are equal — a square. The lesson lets the learner discover this by moving a slider.

Keeping things regulated and calm

- Keep to one clear step at a time; there is no time pressure and no benefit to speed.
- Treat a wrong or unfinished answer as information for the next step, never as failure.
- Let the learner choose how to record — typed, written, drawn, spoken or annotated.
- If frustration rises, it is fine to pause and return later; the lesson holds their place.
- Avoid any language about age, school year or pace as a judgement of ability.

Recognising progress

Progress is described as Approaching Proficiency → Developing Proficiency → Proficiency → Refining Proficiency across understanding, application, communication and reflection. These describe where a learner is right now, not labels of what they are. Real understanding shown in part is still real understanding, even while other pieces are still settling.

Where they are	What it can look like
Approaching Proficiency	Beginning to engage; names some features and makes a start with support.

Developing Proficiency	Carries out the method with prompts; can explain part of the reasoning.
Proficiency	Works through independently and explains why each step is true.
Refining Proficiency	Applies the idea confidently to a new or unfamiliar case, and could teach it to someone else.

A note on evidence

Evidence can be captured in whatever form suits the learner — a photo of working, a short spoken explanation, an annotated Scratchpad image, or a written solution. There is no single required format. Incomplete work is information for the next connection, never a failure: it tells you exactly where to begin next time. The aim is a true record of the learner's thinking, not a tidy performance.

If a learner is stuck or distressed

Slow the pace and reduce the load: return to one concrete, familiar case and a single small question. Offer a Reconnection Route. Remind them they can pause and come back. A short, successful, low-stakes step rebuilds confidence more reliably than pushing through. Connection before curriculum — always.